VALIDATION PLAN
OPERATIONAL QUALIFICATION
RELIABILITY DEMONSTRATION TESTING MODULE

JR Validated Environment — Reliability Demonstration Testing Module v1.0

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Version History

Version	Date	Author	Description
1.0	2026-04-22	Joep Rous	Initial release. Covers both RDT module scripts (jrc_rdt_plan, jrc_rdt_verify) in module v1.0. 25 test cases; all numeric assertions computed independently of R using closed-form Python arithmetic.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-04-22
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document defines the Validation Plan for the Operational Qualification (OQ) of the Reliability Demonstration Testing (RDT) module included in version 1.0 of the JR Validated Environment. It specifies the user requirements, test cases, acceptance criteria, and traceability between requirements and tests needed to demonstrate that each RDT script performs correctly for its intended use in FDA design verification under 21 CFR 820.30(f) and ISO 13485:2016 §7.3.6.

Execution of the tests defined in this plan, together with satisfactory results and documented evidence, constitutes the OQ for the RDT module.

2. Scope

2.1 In Scope

- Both R scripts in repos/rdt/R/ (jrc_rdt_plan.R, jrc_rdt_verify.R)
- Correct computation of required sample size (n) using the Bogey/binomial method (no Weibull shape assumption) and the Weibayes method (Weibull shape β provided)
- Correct effect of acceleration factor (AF) on sample size: n reduces by AF^β in Weibayes mode; Bogey mode is AF-independent
- Correct beta sensitivity table when accel_factor > 1
- Correct Binomial (Clopper-Pearson exact) verification: R_lower = 1 − qbeta(C, k+1, n−k)
- Correct Weibayes verification: R_demo = exp(−(TL/η_demo)^β), where η_demo uses sum(t_eff_i^β)
- Correct overall verdict logic: PASS if Binomial OR Weibayes passes
- Correct rejection of invalid inputs with informative error messages and non-zero exit
- Correct file I/O (CSV input, PNG output to ~/Downloads/)
- Verification that all RDT scripts enforce execution within the jrrun validated environment by checking RENV_PATHS_ROOT at startup

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by IQ; see JR-VP-001)
- Performance qualification (PQ) — conducted separately under specific test protocols
- User acceptance testing (UAT)
- Validation of R language interpreter and third-party packages (ggplot2) — package integrity is confirmed at installation via SHA256 (JR-IQ-001)
- Statistical validation of the underlying methods (Clopper-Pearson, Weibayes chi-squared pivot) — method references are cited in Section 3
- Weibull shape parameter estimation (β) — covered by jrc_weibull (separate script)

3. Regulatory Basis and References

3.1 Regulations

Reference	Title
21 CFR Part 820, §820.30(f)	Design Verification
21 CFR Part 820, §820.70(i)	Quality System Regulation — Automated Data Processing
21 CFR Part 11	Electronic Records; Electronic Signatures

ISO 13485:2016, §7.3.6	Design and development verification

3.2 FDA Guidance

Reference	Title	Date
FDA GPSV	General Principles of Software Validation; Final Guidance for Industry and FDA Staff	January 2002
FDA CSA	Computer Software Assurance for Production and Quality System Software; Draft Guidance	September 2022

3.3 Statistical Methods

Reference	Title
Meeker, Hahn & Escobar (2017)	Statistical Intervals: A Guide for Practitioners and Researchers, 2nd ed. Wiley. Chapter 8 — Weibayes chi-squared pivot for reliability demonstration testing.
Nelson (2004)	Accelerated Testing: Statistical Models, Test Plans, and Data Analysis. Wiley. Weibayes method with known Weibull shape.
Clopper & Pearson (1934)	The use of confidence or fiducial limits illustrated in the case of the binomial. Biometrika 26(4), 404–413. Exact one-sided confidence bound on failure fraction: F_upper = qbeta(C, k+1, n−k).

3.4 Internal Documents

Document Number	Title
JR-VP-001	Validation Plan — Installation Qualification, JR Validated Environment
JR-IQ-001	IQ Execution Evidence (docs/IQ_validation_*.txt)
JR-VP-002	Validation Plan — OQ, Community Script Suite

4. Definitions and Abbreviations

Term	Definition
RDT	Reliability Demonstration Test. A statistically designed experiment to demonstrate that a device meets a pre-specified reliability claim (e.g. R ≥ 0.95 at target life, 90% confidence).
R (reliability)	The proportion of units that survive to target life under use conditions. Distinct from R (gas constant) used in the Arrhenius equation.
C (confidence)	Statistical confidence level for the reliability claim (e.g. 0.90 = 90%).
k	Number of allowed (or observed) failures at or before target life.
n	Required (or actual) number of test units.
AF (accel_factor)	Life-extension multiplier. t_eff = AF × target_life per unit. AF > 1 means units are tested beyond target life, reducing n in Weibayes mode.
β (beta)	Weibull shape parameter. Governs the failure rate profile: β = 1 (random), β > 1 (wear-out), β < 1 (early failures). Required for Weibayes mode.
Bogey / Binomial	RDT method that makes no Weibull shape assumption. n is independent of AF. Exact binomial criterion: n = ⌈qchisq(C, 2(k+1)) / (2 × (−ln R))⌉. At k=0 this equals the zero-failure success-run formula.
Weibayes	RDT method using an assumed Weibull shape β. Incorporates AF: n = ⌈qchisq(C, 2(k+1)) / (2 × (−ln R) × AF^β)⌉.
Clopper-Pearson	Exact one-sided binomial confidence interval. F_upper = qbeta(C, k+1, n−k); R_lower = 1 − F_upper. PASS if R_lower ≥ reliability_claim.
T*	Weibayes accumulated Weibull time: T* = Σ(t_eff_i^β) over all n units.
T_threshold	Minimum T* required to demonstrate the claim: T_threshold = qchisq(C, 2(k+1)) × TL^β / (2 × (−ln R)).
IQ	Installation Qualification
OQ	Operational Qualification
UR	User Requirement
TC	Test Case

jrrun	The validated bash entry-point script that sets the controlled environment before executing any script
PNG	Portable Network Graphics output file; plots saved to ~/Downloads/

5. User Requirements

The following user requirements define the expected behaviour of the RDT module. Each requirement is verifiable by the test cases in Section 7.

UR ID	Title	Requirement
UR-RDT-001	Bypass protection — RENV_PATHS_ROOT	All RDT scripts shall check the RENV_PATHS_ROOT environment variable at startup and call stop() with an informative message if it is not set, preventing execution outside the validated jrrun environment.
UR-RDT-002	Error handling — non-zero exit	All RDT scripts shall exit with a non-zero status code whenever an invalid input is detected (out-of-range parameter, missing required argument, bad CSV), and shall print an informative error message identifying the cause.
UR-RDT-003	Plan — Bogey/binomial sample size	jrc_rdt_plan shall compute required sample size for k = 0 to 5 allowed failures using the exact Bogey/binomial formula: n = ⌈qchisq(C, 2(k+1)) / (2 × (−ln R))⌉. At k=0, C=0.90, R=0.95 the result shall be n = 45.
UR-RDT-004	Plan — Weibayes sample size with acceleration factor	When --beta is provided, jrc_rdt_plan shall compute n using the Weibayes formula: n = ⌈qchisq(C, 2(k+1)) / (2 × (−ln R) × AF^β)⌉. At AF=2, β=2, k=0, C=0.90, R=0.95 the result shall be n = 12. At AF=1 the result is identical to Bogey regardless of β.
UR-RDT-005	Plan — beta sensitivity table	When --beta is provided and accel_factor > 1, jrc_rdt_plan shall display a beta sensitivity table showing n for a grid of beta values (β × [0.5, 0.75, 1.0, 1.25, 1.5], minimum 0.5) at k=0. At accel_factor = 1 the table shall note that n is beta-independent.
UR-RDT-006	Plan — k_allowed flag	jrc_rdt_plan shall accept --k_allowed K (non-negative integer, default 0). The row corresponding to k = K shall be marked as the plan row in the output table.
UR-RDT-007	Plan — PNG output	jrc_rdt_plan shall save a two-panel PNG to ~/Downloads/ with a timestamp prefix and the suffix _jrc_rdt_plan.png. Panel 1: bar chart of n vs k. Panel 2: beta sensitivity line plot (when applicable).
UR-RDT-008	Verify — CSV input validation	jrc_rdt_verify shall accept a CSV file containing at least a time column and a status column (default names 'time' and 'status'; overridable via --time_col and --status_col). All time values must be positive numerics. Status values must be 0 (survived) or 1 (failed). Missing or wrong-named columns shall cause a non-zero exit.
UR-RDT-009	Verify — Binomial (Clopper-Pearson) verification	jrc_rdt_verify shall compute the exact Clopper-Pearson upper confidence bound: F_upper = qbeta(C, k+1, n−k), where k = failures with t_eff ≤ target_life. R_lower = 1 − F_upper. When n−k = 0 (all units failed), F_upper = 1 and R_lower = 0. PASS if R_lower ≥ reliability_claim. At n=45, k=0, C=0.90 the result shall be R_lower ≈ 0.9501.
UR-RDT-010	Verify — Weibayes verification	When --beta is provided, jrc_rdt_verify shall compute the Weibayes demonstrated reliability: T* = Σ(t_eff_i^β); η_demo = (2T* / qchisq(C, 2(k+1)))^(1/β); R_demo = exp(−(TL/η_demo)^β). PASS if R_demo ≥ reliability_claim. At n=45, β=2, t=5000, k=0, C=0.90 the result shall be R_demo ≈ 0.9501.
UR-RDT-011	Verify — overall verdict	jrc_rdt_verify shall report an overall verdict of PASS if either the Binomial or the Weibayes method (when applicable) produces a PASS result. When the two methods disagree, a note shall be printed.

UR-RDT-012	Verify — exit code convention	jrc_rdt_verify shall exit 0 for both PASS and FAIL verdicts. Non-zero exit is reserved for input errors and runtime failures.
UR-RDT-013	Verify — PNG output	jrc_rdt_verify shall save a two-panel PNG to ~/Downloads/ with a timestamp prefix and the suffix _jrc_rdt_verify.png. Panel 1: unit timeline (survivors = +, failures = ×). Panel 2: demonstrated reliability bars (Binomial and Weibayes if applicable).

6. Test Environment

Component	Specification
Operating System	macOS 15.x (Apple Silicon) or Windows 10/11 (Git Bash)
R Version	As installed per JR-IQ-001 (R 4.3.x or later)
Python Version	As installed per JR-IQ-001 (Python 3.11.x or later)
Test Framework	pytest 8.x
Test runner	repos/rdt/admin_rdt_oq
Test files	repos/rdt/oq/test_rdt_plan.py (13 tests), repos/rdt/oq/test_rdt_verify.py (12 tests)
Test data	repos/rdt/oq/data/ (5 CSV fixtures)
OQ venv	~/.venvs/<PROJECT_ID>_oq/ (shared with other module OQ suites)
Output directory	~/Downloads/ (PNG files verified by mtime glob)
Entry point	bin/jrrun (integrity-checked execution)

7. Test Cases

All test cases are implemented in repos/rdt/oq/test_rdt_plan.py and repos/rdt/oq/test_rdt_verify.py. Test data files reside in repos/rdt/oq/data/. Numerical reference values are computed independently of the R scripts using pure Python arithmetic (math.log, math.exp, math.ceil, bisection on exact CDFs) — no R or scipy is used in the reference calculations.

7.1 jrc_rdt_plan

TC-RDT-PLAN-001

Tests: UR-RDT-003
Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000`
Rationale: Bogey mode (no --beta): verifies the mode header appears in output.
Pass criterion: exit 0; 'Binomial' in output

TC-RDT-PLAN-002

Tests: UR-RDT-004
Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000 --beta 2.0`
Rationale: Weibayes mode: verifies the mode header appears in output.
Pass criterion: exit 0; 'Weibayes' in output

TC-RDT-PLAN-003

Tests: UR-RDT-003
Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000`
Rationale: Known numeric, Bogey k=0. Reference: n = ceil(log(0.10)/log(0.95)) = 45. Computed in pure Python (math.ceil, math.log), independent of R.
Pass criterion: exit 0; 'n = 45' in output

TC-RDT-PLAN-004

Tests: UR-RDT-004

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000 --beta 2.0`

Rationale: Weibayes k=0 AF=1: AF^β = 1^2 = 1 for any β. Formula reduces to Bogey → n=45.

Pass criterion: exit 0; 'n = 45' in output

TC-RDT-PLAN-005

Tests: UR-RDT-004

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000 --beta 2.0 --accel_factor 2.0`

Rationale: Weibayes k=0, β=2, AF=2. Reference: n = ceil(log(0.10)/(log(0.95)*4)) = 12. Computed in pure Python, independent of R.

Pass criterion: exit 0; 'n = 12' in output

TC-RDT-PLAN-006

Tests: UR-RDT-003, UR-RDT-006

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000 --k_allowed 1`

Rationale: k=1 Bogey. Reference: qchisq(0.90, 4) solved via bisection on exact Erlang-2 CDF in pure Python; n = ceil(7.7794 / 0.10258) = 76. Plan marker '<- plan' on k=1 row.

Pass criterion: exit 0; 'n = 76' in output; '<- plan' in output

TC-RDT-PLAN-007

Tests: UR-RDT-002

Command: `jrrun jrc_rdt_plan.R --help`

Rationale: --help flag: exits 0 with usage text.

Pass criterion: exit 0; 'Usage' in output

TC-RDT-PLAN-008

Tests: UR-RDT-005

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000 --beta 2.0 --accel_factor 2.0`

Rationale: Beta sensitivity table: requires --beta and accel_factor > 1.

Pass criterion: exit 0; 'Beta Sensitivity' in output

TC-RDT-PLAN-009

Tests: UR-RDT-007

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000`

Rationale: PNG output to ~/Downloads/.

Pass criterion: exit 0; *_jrc_rdt_plan.png found in ~/Downloads/ after run

TC-RDT-PLAN-010

Tests: UR-RDT-002

Command: `jrrun jrc_rdt_plan.R --reliability 1.5 --confidence 0.90 --target_life 5000`

Rationale: --reliability = 1.5 (outside (0,1)).

Pass criterion: exit non-zero

TC-RDT-PLAN-011

Tests: UR-RDT-002

Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --confidence 0 --target_life 5000`

Rationale: --confidence = 0 (outside (0,1)).

Pass criterion: exit non-zero

TC-RDT-PLAN-012

Tests: UR-RDT-002
Command: `jrrun jrc_rdt_plan.R --reliability 0.95 --target_life 5000`
Rationale: Missing required argument --confidence.
Pass criterion: exit non-zero

TC-RDT-PLAN-013

Tests: UR-RDT-001
Command: `Rscript repos/rdt/R/jrc_rdt_plan.R --reliability 0.95 --confidence 0.90 --target_life 5000  (RENV_PATHS_ROOT unset)`
Rationale: Direct Rscript call bypasses jrrun.
Pass criterion: exit non-zero; 'RENV_PATHS_ROOT' in output

7.2 jrc_rdt_verify

TC-RDT-VER-001

Tests: UR-RDT-008, UR-RDT-009, UR-RDT-012
Command: `jrrun jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000`
Rationale: rdt_verify_pass.csv: 45 units, all survived (status=0, t=5000). Happy path.
Pass criterion: exit 0; 'Verdict' in output

TC-RDT-VER-002

Tests: UR-RDT-009, UR-RDT-012
Command: `jrrun jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000`
Rationale: Numeric assertion: n=45, k=0. R_lower = 1 − (1−0.90)^(1/45) ≈ 0.9501. Reference computed in pure Python (exact Beta(1,n) ppf). Tolerance ±0.0002.
Pass criterion: exit 0; 'PASS' in output; R_lower = 0.9501 ± 0.0002

TC-RDT-VER-003

Tests: UR-RDT-009, UR-RDT-012
Command: `jrrun jrc_rdt_verify.R rdt_verify_fail.csv --reliability 0.95 --confidence 0.90 --target_life 5000`
Rationale: rdt_verify_fail.csv: 20 units all survived. n=20, k=0. R_lower = 1 − 0.10^(1/20) ≈ 0.8913 < 0.95. FAIL verdict; script exits 0 (FAIL is not a script error).
Pass criterion: exit 0; 'FAIL' in output; R_lower = 0.8913 ± 0.0002

TC-RDT-VER-004

Tests: UR-RDT-010, UR-RDT-011
Command: `jrrun jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000 --beta 2.0`
Rationale: Weibayes with n=45, β=2, t=5000, k=0. Reference: $T^* = 45\times5000^2 = 1.125\times10^9$; qchisq(0.90,2) = −2×ln(0.10) ≈ 4.6052 (exact). η_demo = (2×T*/4.6052)^0.5 ≈ 22104; R_demo = exp(−(5000/22104)²) ≈ 0.9501. Tolerance ±0.0002.
Pass criterion: exit 0; 'Weibayes' in output; 'PASS' in output; R_demo = 0.9501 ± 0.0002

TC-RDT-VER-005

Tests: UR-RDT-013
Command: `jrrun jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000`
Rationale: PNG output to ~/Downloads/.
Pass criterion: exit 0; *_jrc_rdt_verify.png found in ~/Downloads/ after run

TC-RDT-VER-006

Tests: UR-RDT-009
Command: `jrrun jrc_rdt_verify.R rdt_verify_all_failed_early.csv --reliability 0.95 --confidence 0.90 --target_life 5000`
Rationale: rdt_verify_all_failed_early.csv: 10 units, all failed at t=2500 < target_life=5000. Edge case: k=n=10, n−k=0. Script sets F_upper=1.0, R_lower=0.0. Verified by extract on 'R_lower =' label.
Pass criterion: exit 0; 'FAIL' in output; R_lower < 0.01

TC-RDT-VER-007

Tests: UR-RDT-009
Command: `jrrun jrc_rdt_verify.R rdt_verify_zero_failures.csv --reliability 0.90 --confidence 0.90 --target_life 5000`
Rationale: rdt_verify_zero_failures.csv: 50 units, all survived. R_lower = 1 − 0.10^(1/50) ≈ 0.9550. Reference computed in pure Python. Assert R_lower ∈ [0.953, 0.959].
Pass criterion: exit 0; R_lower in [0.953, 0.959]

TC-RDT-VER-008

Tests: UR-RDT-002
Command: `jrrun jrc_rdt_verify.R  (no arguments)`
Rationale: --help shown when no arguments provided; exits 0.
Pass criterion: exit 0; 'Usage' in output

TC-RDT-VER-009

Tests: UR-RDT-008, UR-RDT-002
Command: `jrrun jrc_rdt_verify.R /tmp/no_such_file_rdt_oq.csv --reliability 0.95 --confidence 0.90 --target_life 5000`
Rationale: Non-existent file path.
Pass criterion: exit non-zero; 'not found' or 'File' in output

TC-RDT-VER-010

Tests: UR-RDT-008
Command: `jrrun jrc_rdt_verify.R rdt_verify_missing_col.csv --reliability 0.95 --confidence 0.90 --target_life 5000`
Rationale: rdt_verify_missing_col.csv has column 'hours' not 'time'. Error message must name the missing column.
Pass criterion: exit non-zero; 'time' in error output

TC-RDT-VER-011

Tests: UR-RDT-008
Command: `jrrun jrc_rdt_verify.R <inline temp CSV with non-numeric time> --reliability 0.95 --confidence 0.90 --target_life 5000`
Rationale: Temp CSV created in-test with time='abc'. Non-numeric time value.
Pass criterion: exit non-zero

TC-RDT-VER-012

Tests: UR-RDT-001
Command: `Rscript repos/rdt/R/jrc_rdt_verify.R rdt_verify_pass.csv --reliability 0.95 --confidence 0.90 --target_life 5000  (RENV_PATHS_ROOT unset)`
Rationale: Direct Rscript call bypasses jrrun.
Pass criterion: exit non-zero; 'RENV_PATHS_ROOT' in output

8. Requirements Traceability Matrix

Each user requirement is covered by at least one test case.

UR ID	Title	Test Cases
UR-RDT-001	Bypass protection (RENV_PATHS_ROOT)	TC-RDT-PLAN-013, TC-RDT-VER-012
UR-RDT-002	Non-zero exit on error	TC-RDT-PLAN-007, TC-RDT-PLAN-010, TC-RDT-PLAN-011, TC-RDT-PLAN-012, TC-RDT-VER-008, TC-RDT-VER-009, TC-RDT-VER-010, TC-RDT-VER-011
UR-RDT-003	Plan — Bogey/binomial sample size	TC-RDT-PLAN-001, TC-RDT-PLAN-003, TC-RDT-PLAN-006
UR-RDT-004	Plan — Weibayes sample size with AF	TC-RDT-PLAN-002, TC-RDT-PLAN-004, TC-RDT-PLAN-005
UR-RDT-005	Plan — beta sensitivity table	TC-RDT-PLAN-008
UR-RDT-006	Plan — k_allowed flag	TC-RDT-PLAN-006
UR-RDT-007	Plan — PNG output	TC-RDT-PLAN-009
UR-RDT-008	Verify — CSV input validation	TC-RDT-VER-001, TC-RDT-VER-009, TC-RDT-VER-010, TC-RDT-VER-011
UR-RDT-009	Verify — Binomial (Clopper-Pearson)	TC-RDT-VER-001, TC-RDT-VER-002, TC-RDT-VER-003, TC-RDT-VER-006, TC-RDT-VER-007
UR-RDT-010	Verify — Weibayes verification	TC-RDT-VER-004
UR-RDT-011	Verify — overall verdict	TC-RDT-VER-004
UR-RDT-012	Verify — exit code convention	TC-RDT-VER-001, TC-RDT-VER-002, TC-RDT-VER-003, TC-RDT-VER-008, TC-RDT-VER-009
UR-RDT-013	Verify — PNG output	TC-RDT-VER-005

9. Acceptance Criteria

The RDT module OQ is considered PASS when all of the following conditions are satisfied:

- All 25 automated test cases execute with the result PASSED.
- Zero test cases result in FAILED or ERROR.
- The OQ evidence file is generated and preserved at ~/.jrscript/<PROJECT_ID>/validation/rdt_oq_execution_<timestamp>.txt.
- All test cases are executed using the validated jrrun entry-point (bypass protection verified for both scripts).
- No open deviations remain unresolved at the time of approval.

Any test failure shall be investigated, documented as a deviation, resolved, and the affected test(s) re-executed. The OQ may not be closed until all deviations are resolved.